

// Day 2 of 10

Day 2

10 SQL Mistakes

Developers make these every day.
Don't be one of them.

→ Swipe through all 10



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Mistake #01

SELECT * Everywhere

Fetches unused columns, kills speed

// Bad

```
SELECT * FROM users;
```

// Good

```
SELECT id, name, email  
FROM users;
```



Always select only needed columns



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Mistake #02

No WHERE on UPDATE/DELETE

One typo = entire table wiped

```
// Bad
```

```
UPDATE users SET status = 'inactive';  
-- Updates ALL rows!
```

```
// Good
```

```
UPDATE users SET status = 'inactive'  
WHERE last_login < '2023-01-01';
```



Always use WHERE on UPDATE/DELETE



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Mistake #03

No Indexes on Joins

Full table scan = very slow queries

```
// Bad
-- No index on foreign key
SELECT * FROM orders o
JOIN users u ON o.user_id = u.id
```

```
// Good
CREATE INDEX idx_orders_user
  ON orders(user_id);
-- Now joins are fast
```



Index all foreign key columns



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Mistake #04

Using OR Instead of IN

OR chains are harder to optimize

```
// Bad
```

```
WHERE city='Delhi'
```

```
OR city='Mumbai'
```

```
OR city='Pune'
```

```
// Good
```

```
WHERE city IN
```

```
('Delhi', 'Mumbai', 'Pune')
```



IN is cleaner and often faster



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Mistake #05

Skipping LIMIT

One query can crash your DB

// Bad

```
SELECT * FROM logs  
ORDER BY created_at DESC;  
-- Returns millions of rows!
```

// Good

```
SELECT * FROM logs  
ORDER BY created_at DESC  
LIMIT 100;
```



Always LIMIT large result queries



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Mistake #06

Storing Passwords Plain Text

Worst security mistake ever

// Bad

```
INSERT INTO users  
VALUES ('john@test.com', 'pass123');
```

// Good

```
-- Use bcrypt hash before storing  
INSERT INTO users  
VALUES ('john@test.com', '$2b$10$xyz ...');
```



Hash passwords before storing



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Mistake #07

Wrong NULL Comparisons

NULL = NULL is always false in SQL

```
// Bad
```

```
WHERE email = NULL;  
-- This NEVER returns any rows!
```

```
// Good
```

```
WHERE email IS NULL;  
WHERE email IS NOT NULL;
```



Always use IS NULL / IS NOT NULL



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Mistake #08

No Transactions

Partial data = corrupted records

```
// Bad
```

```
INSERT INTO orders ... ;  
INSERT INTO payments ... ; -- crash?  
-- Orders saved, payment not!
```

```
// Good
```

```
BEGIN;  
  INSERT INTO orders ... ;  
  INSERT INTO payments ... ;
```



Wrap related ops in transactions



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Mistake #09

Ignoring EXPLAIN

You can't fix what you can't see

```
// Bad
-- Just running slow queries
-- and hoping for the best
```

```
// Good
EXPLAIN ANALYZE
SELECT * FROM orders
WHERE user_id = 42;
-- See actual query plan!
```



EXPLAIN is your best debug tool



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Mistake #10

Wrong Data Types

Wastes storage and slows queries

// Bad

```
age VARCHAR(10),  
price VARCHAR(20),  
active VARCHAR(5)
```

// Good

```
age INT,  
price DECIMAL(10,2),  
active BOOLEAN,
```



Right type = right performance



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done = true

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